

MULTIREACTION APPROACH TO QUANTUM THERMOCHEMISTRYIrikura K.K.National Institute of Standards and Technology
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Gas-phase enthalpies of formation, $\Delta_f H_0^\circ(X)$, are often predicted using *ab initio* calculations. One selects a balanced chemical reaction involving n species, including the target molecule X . For all species besides X , $(n-1)$ enthalpies of formation are taken from the literature. The reaction enthalpy is then obtained from calculation and the desired quantity is obtained by simple arithmetic. The best accuracy is obtained when the selected reaction successfully cancels the systematic errors in the *ab initio* method.

In practice, there are often difficulties. A clever reaction (e.g., one that conserves Benson groups [1]) may require data that do not exist. Data that do exist may have large uncertainties. Different choices of reaction yield different results. It may be difficult to estimate the uncertainty associated with the thermochemical prediction.

We propose [2] to solve these problems by using more than one reaction in combination with an estimator for the unreliability of each prediction. The unreliability of a reaction may be estimated as proportional to its enthalpy change. The unreliability of a value of $\Delta_f H_0^\circ(X)$ may be estimated as the reaction unreliability plus the uncertainty in the literature data plus a constant representing the unreliability in the *ab initio* method. The values of $\Delta_f H_0^\circ(X)$ obtained from many reactions may be combined using weighted statistics, where the weight of a prediction is the reciprocal of its unreliability. This yields a best estimate for the desired quantity, along with a quantitative estimate of its uncertainty.

1. S. W. Benson and J. H. Buss, J. Chem. Phys. 29, 546–72, (1958).
<https://doi.org/10.1063/1.1744539>
2. K. K. Irikura, J. Phys. Chem. A 124, 8088–99, (2020).
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